

Sprint 3 Simulator Test — Quick Guide

Notebook: Sprint3_Simulator_Test.ipynb
April 9, 2026

1. What This Notebook Does

This notebook loads **simulator.py** and runs it through three mandatory validation gates, then demonstrates scenario runs so you can inspect per-tract results. It also runs the full 16-test pytest suite (**test_simulator.py**) inline.

The simulator is a general-purpose engine — it accepts any combination of lever deltas and tract filters. The scenarios in this notebook are demos, not hardcoded behavior. Sprint 3.3 and the Sprint 4 dashboard will call the same **sim.run()** API with whatever parameters the client chooses.

2. Setup

The notebook must run from the **Sprint 3/** folder. It imports **simulator.py** from the same directory, which in turn loads three files:

File	Location (relative to Sprint 3/)	Loaded by
Sprint3_Baseline_State.csv	Sprint 3/ (same folder)	simulator.py
Sprint2b_XGBoost_v3.pkl	Sprint 2 / (one level up, then into Sprint 2)	simulator.py
Sprint3_Lever_Catalog.json	Sprint 3/ (same folder)	simulator.py

Dependencies: xgboost, joblib, scikit-learn, pandas, numpy, pytest

To run: Open in VS Code, select Python kernel, Run All. Make sure your working directory is Sprint 3/ (VS Code usually sets this to the notebook's folder).

3. How to Read Each Output

Section 1 — Load Simulator

Confirms the simulator loaded successfully. You should see **504 tracts** and **4 levers** (freq_peak_am_tph, freq_early_tph, weekend_weekday_ratio, rail_trip_share). Also shows a 5-row baseline snapshot with equity scores and tiers.

Section 2 — Gate 1: Zero-Delta

Runs the simulator with **no changes**. Every line should say PASS. This proves the proportional-change math is self-consistent — when nothing changes, equity scores don't drift. Expected output:

```
Max |equity_delta|: ~1e-17 PASS
Max |deficit_delta|: 0.00e+00 PASS
Tier shifts: 0 PASS
Summary improved: 0 | worsened: 0
```

Section 3 — Gate 2: Isolation

Applies +2 trips/hr peak frequency to **Critical tracts only** (51 tracts). Checks that the other 453 tracts have exactly zero change. All lines should say PASS. This validates that the tract filter works and that derived features (headway) don't leak to unfiltered tracts.

Section 4 — Gate 3: Direction Tests

Runs each lever independently and checks that the average deficit **decreases**. All four should say PASS. The table also shows tracts improved and tier upgrades:

Scenario	Avg Deficit Δ	Tracts Improved	Tier Upgrades
S1: Peak freq +2 tph	-0.026	494/504	74
S2: Weekend ratio +0.1	-0.007	412/504	23
S3: Early freq +1 tph	-0.004	309/504	14
S4: Rail share +10pp	-0.017	487/504	49

Also checks SHAP ranking consistency: $|S1| > |S3|$ (peak freq is SHAP #1, early freq is SHAP #3 — the bigger lever should have a bigger effect).

Section 5 — Tier Shift Matrices

A 4x4 matrix for each scenario. **Rows** = original tier, **columns** = new tier. Diagonal = tracts that stayed. Off-diagonal = tracts that moved. Example reading: in S1, 16 Critical tracts dropped to High and 27 Moderate tracts dropped to Low. Zero downgrades in S1 and S4 (pure improvement). S2 and S3 show a few downgrades (some tracts worsen slightly when weekend ratio or early freq changes).

Section 6 — Per-Tract Detail (S4 Rail Shift)

Shows the 10 most improved tracts and any worsened tracts for the rail modal shift scenario. Tract 12086008904 improves most (-0.048 equity). Only 2 tracts worsen, both by negligible amounts ($\sim 1e-17$ to $3e-4$). Also shows fragile tract impacts: 16 of 22 fragile tracts improved, avg equity drop of -0.008.

Section 7 — Pytest (16 tests)

Runs `test_simulator.py` inline. You should see **16 passed** in green. This is the same test suite you'd run from terminal with `pytest test_simulator.py -v`. The 1 warning about an escape sequence is cosmetic and harmless.

Section 8 — Combined Multi-Lever Demo

Combines 3 levers (peak freq +2, rail +10pp, weekend +0.1) in one run. Shows that the combined effect (-0.052 avg deficit) is larger than any single lever alone, with 502/504 tracts improved and 148 tier upgrades. This previews Sprint 3.3 Scenario S5.

4. File Reference

File	Role
<code>simulator.py</code>	The simulation engine. General-purpose, no hardcoded scenarios.
<code>test_simulator.py</code>	16 pytest validation tests (developer sanity checks).
<code>Sprint3_Simulator_Test.ipynb</code>	This notebook — interactive demo + validation.
<code>Sprint3_Baseline_State.csv</code>	504 tracts: features + equity + need projections.
<code>Sprint3_Lever_Catalog.json</code>	4 lever definitions with ranges and SHAP info.

Sprint2b_XGBoost_v3.pkl	Trained XGBoost model (loaded by simulator.py).
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